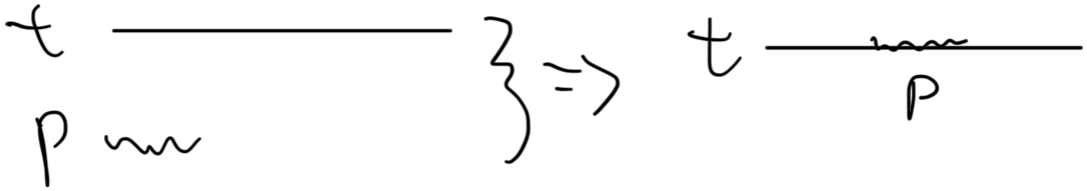
10/29 CSCI 1810 Notes

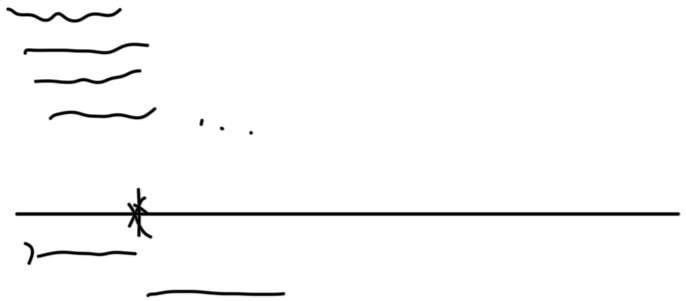
Knuth-Morris-Pratt Algorithm $O(n+l)$

Problem: Given a pattern $p = p_1 p_2 \cdots p_l$
and a text $t = t_1 t_2 \cdots t_n$

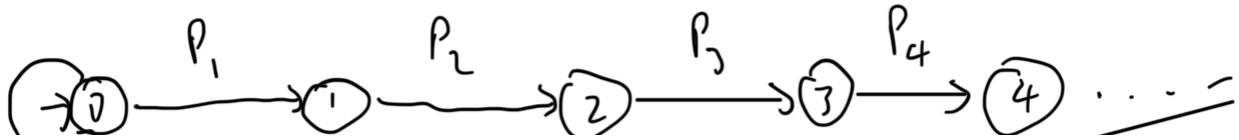
Compute: the first exact occurrence of p in t



$O(n \times l)$ too slow

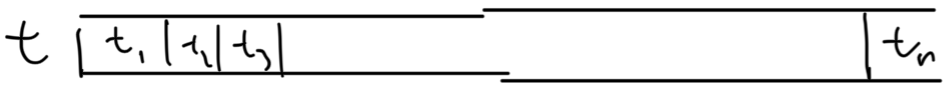


Skeleton DFA M_p: p defines the DFA





Input Tape



when a mismatch occurs $t_{k+1} \neq p_i t_i$
then M_p enters the highest number
state (i) such that $p_1 p_2 \cdots p_i$ is a suffix of
$t_1 t_2 \cdots t_{k+1}$
$p_1 p_2 \cdots p_i$ => basically the prefix is equal to the suffix

Algo for computing Failure Function : f of p

An Example: $p = \overset{1}{a}\overset{2}{a}\overset{3}{b}\overset{4}{b}\overset{5}{a}\overset{6}{a}\overset{7}{b}$

the failure func. for p

i	1	2	3	4	5	6	7
f(i)	0	1	0	0	1	2	3

To help determine i, the machine M_p has associated with the ... value function

[illegible] with it an integer

f called the failure function such that $f(j)$ is the largest s less than j for which $p_1 \cdots p_s$ is a suffix of $p_1 \cdots p_j$

$p_1 \cdots p_s = p_{j-s+1} \cdots p_{j-1} p_j$

If there is no such $s \geq 1$, then $f(j) = 0$

$f(7) = 3$ since aab is the longest proper prefix of aabbaab

We will present an algorithm to compute the failure function

To see how it's used by M_p, let us define the following function $f^{(m)}(j)$ as follows

i) $f^{(1)}(j) = f(j)$

ii) $f^{(m)}(j) = f(f^{(m-1)}(j))$ for $m > 1$

that is $f^{(m)}(j)$ = f applied "m" times to j

$f^{(2)}(j) = f(f(j)) = f(f(f(j))) = f(2) = 1$

Ex. $f^*(6) = f(f(0), f(1)) = 1$

DFA { M_p is nondeterministic
f
$\Rightarrow$ combine to make a DFA

Resuming from when we was stuck at $t_{k+1} \neq P_{j+1}$ ↓

Suppose that M_p is in a state (j) having read $t_1 t_2 \ldots t_k$ on the input tape and $t_{k+1} \neq P_{j+1}$, M_p applies the failure function f repeatedly to j until it finds th smallest value m for which either...

Case 1: $f^{(m)}(j) = u$ and $t_{k+1} = P_{u+1}$,
happy case! You can continue

OR

Case 2: $f^{(m)}(j) = 0$ and $t_{k+1} \neq P_1$

That is, M_p backs up through states $f^{(2)}(j)$, $f^{(3)}(j)$, ... until we get to case 1 or case 2 holds for $f^{(m)}(j)$ but not for $f^{(m-1)}(j)$

In Case 1, M_p enters state (j+1)

In case 2, M_p enters state (0)

In either case, the tape head advances to th next letter on the tape: t+2

In Case 1, it's easy to see that $P_1 P_2 \ldots P_j$ was the longest prefix of p that is a suffix of $t_1 t_2 \ldots t_k$ then $P_1 P_2 \ldots P_{f(j)+1}$ is the longest prefix of p that is a suffix of $t_1 t_2 \ldots t_k t_{k+1}$.

In can 2, no prefix of p is a suffix of $t_1 t_2 \ldots t_k t_{k+1}$

Next, M_p proceeds processing input tape symbol $t_i t_{i+1}$, and so on and so on...

M_p continues this until it enters final state (l) in which case p occurs exactly in t $\hookrightarrow p_1 p_2 \cdots p_\ell$

OR

M_p continues and reaches the end of the text and never enters state (l) in which case, the conclusion is p doesn't occur exactly in t

---

The Failure Function Algorithm

Input: $P = p_1 p_2 \cdots p_\ell \quad \ell \geq 1$

Output: f the Failure Function of P

Begin

L_1 --- $f(1) := 0$; $i := 0$;

L_2 --- For $j := 2$ until ℓ Do

```
                    begin
L3 . ~ - - - - ~~ - i := f(j-1)
L4 .  -  -  -  . - while P_j ≠ P_(i+1) And i > 0
                       Do i := f(i);
L5 . -  -  -  - If P_j ≠ P_(i+1) AND i = 0
                       Then f(j) = 0
L6 .  ~  -  -  - - ELSE f(j) = i+1
                  END
```